Exercise Sheet 5

The problems use complex finite-dimensional spaces. Casimir operators are formed with the trace form of the module, as in Lecture 5. Exercises 5.1–5.3 compute Casimir operators and compare forms. Exercise 5.4 treats isotypic multiplicities and commutants. Exercises 5.5–5.6 locate the failure of complete reducibility outside the semisimple case.

Exercise 5.1 (The defining $\mathfrak{sl}_n$-module). Let $n \geq 2$ and let $\mathfrak{sl}_n$ act on $V = \mathbb{C}^n$ by matrix multiplication. Its trace form is $\beta_V(X, Y) = \operatorname{tr}(XY)$.

(i) Prove that V is irreducible.

(ii) Deduce that $c_V = \frac{n^2-1}{n} I_n$.

(iii) Verify (ii) directly. Use the basis consisting of E_{ij} for $i \neq j$ and $h_k = E_{kk} - E_{k+1,k+1}$ for $1 \leq k \leq n-1$. Show that the dual basis consists of E_{ji} and

$$\omega_k = \sum_{i \leq k} E_{ii} - \frac{k}{n} I_n.$$

Exercise 5.2 (Invariant forms on a simple algebra). Let $\mathfrak{g}$ be simple. Let β be a non-degenerate symmetric invariant bilinear form on $\mathfrak{g}$, and let β' be any invariant bilinear form.

(i) Prove that $\beta' = t\beta$ for some $t \in \mathbb{C}$.

(ii) Let V be a module with $\rho \neq 0$. Prove that $\beta_V = t_V \kappa$ with $t_V \neq 0$. If (y_i^κ) is the basis dual to (x_i) for κ, prove that

$$c_V = t_V^{-1} \sum_i \rho(x_i)\rho(y_i^\kappa).$$

Exercise 5.3 (A Casimir operator restricted to a summand). Let $\mathfrak{g} = \mathfrak{sl}_2$, let $U = \mathbb{C}^2$ be the defining module, let A be the adjoint module, and let $V = U \oplus A$.

(i) Compute the Gram matrix of β_U in the basis (h, e, f), and recall the Gram matrix of $\kappa = \beta_A$ from Lecture 4. Deduce $\beta_V = \frac{5}{4}\kappa$.

(ii) Compute c_V on U and on A.

(iii) Compare with c_U and c_A, and check $\operatorname{tr} c_V$.

Exercise 5.4 (Isotypic multiplicities and commutants). Let $\mathfrak{g}$ be semisimple and let V be a module. By Weyl's theorem,

$$V \cong \bigoplus_{i=1}^{r} L_i^{\oplus m_i},$$

where $L_1, \dots, L_r$ are pairwise nonisomorphic irreducible modules and $m_i \geq 1$.

(i) Prove $\dim \operatorname{Hom}_{\mathfrak{g}}(L_j, V) = m_j$. Deduce that the multiplicities do not depend on the chosen decomposition.

(ii) Prove that $\operatorname{End}_{\mathfrak{g}}(V) \cong \prod_{i=1}^{r} M_{m_i}(\mathbb{C})$ as algebras. In particular $\dim \operatorname{End}_{\mathfrak{g}}(V) = \sum_i m_i^2$.

(iii) For $\mathfrak{g} = \mathfrak{sl}_2$, find $\operatorname{End}_{\mathfrak{g}}(V)$ for $V = \mathfrak{gl}_2$ as in Lecture 5, and for $V = \mathbb{C}^2 \oplus \mathbb{C}^2$.

Exercise 5.5 (The Hom-space argument for a solvable algebra). Let $\mathfrak{b} = \mathbb{C}h \oplus \mathbb{C}e$ with $[h, e] = e$, as in Lecture 3. Let $V = \mathbb{C}^2$ have action matrices $D(h) = E_{11}$ and $D(e) = E_{12}$ in the standard basis (v_1, v_2), and put $W = \mathbb{C}v_1$. Write each $\varphi \in \mathrm{Hom}_{\mathbb{C}}(V, W)$ as

$$\varphi(z_1 v_1 + z_2 v_2) = (sz_1 + tz_2)\, v_1,$$

and record it as the row (s, t).

(i) Compute $h \cdot (s, t)$ and $e \cdot (s, t)$.

(ii) Identify $\mathcal{V}$ and $\mathcal{W}$ from the proof of Weyl's theorem, the action on $\mathcal{W}$ and the action on $\mathcal{V}/\mathcal{W}$.

(iii) Prove that $\mathcal{W}$ has no complementary submodule in $\mathcal{V}$, and that $\mathrm{Hom}_{\mathfrak{b}}(V, W) = 0$.

Exercise 5.6 (Central action of $\mathfrak{gl}_n$). Let $n \geq 2$ and $\mathfrak{g} = \mathfrak{gl}_n(\mathbb{C})$. Let $N = E_{12} \in \mathfrak{gl}_2(\mathbb{C})$.

(i) Prove that $\rho(X) = (\mathrm{tr}\, X)\, N$ defines a representation of $\mathfrak{gl}_n$ on $\mathbb{C}^2$ which is not completely reducible.

(ii) Compute the trace form of this module.

(iii) Let V be a $\mathfrak{gl}_n$-module on which $\rho(I_n)$ is diagonalisable. Prove that V is completely reducible.